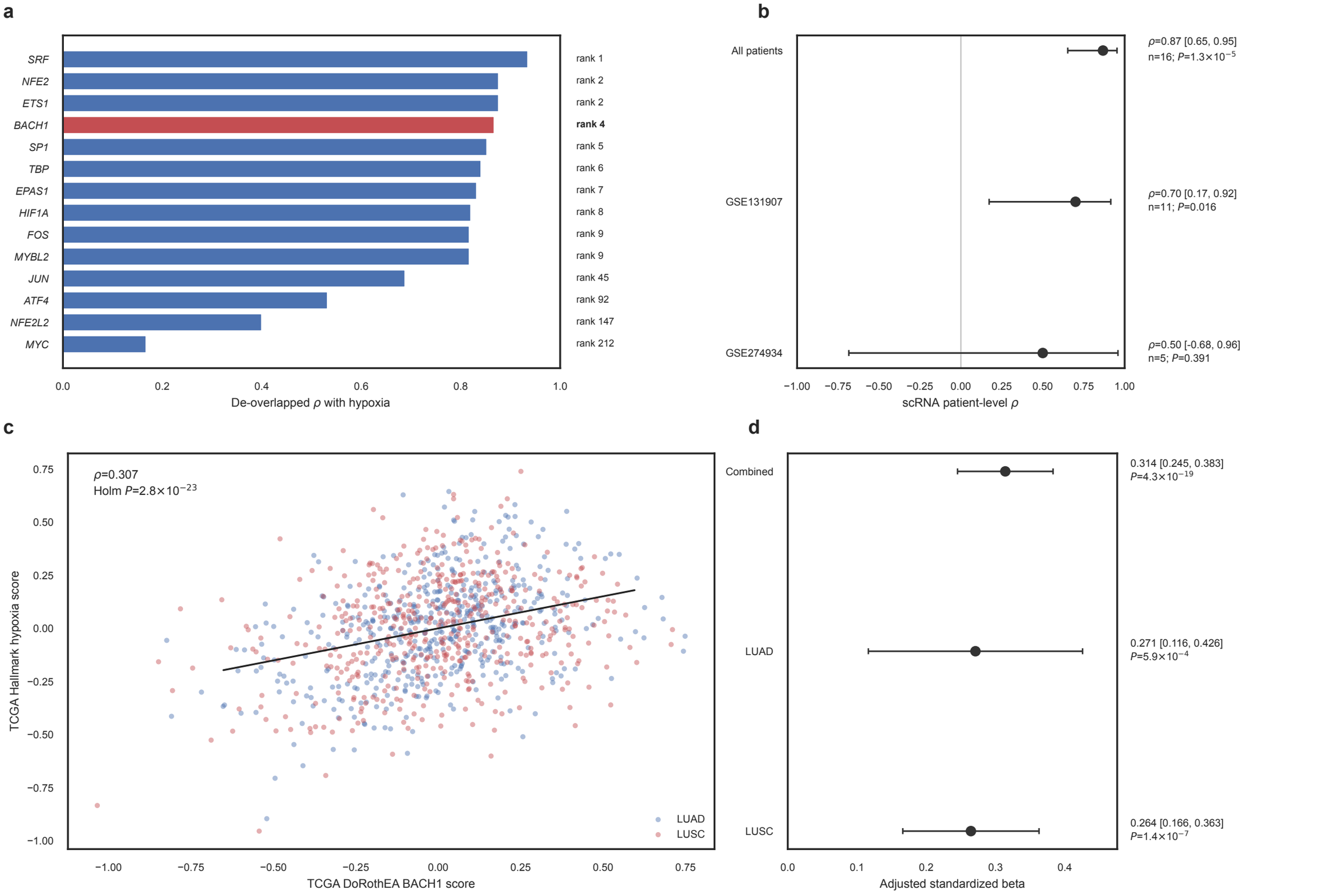

a
SRF
NFE2
ETS1
BACH1
SP1
TBP
EPAS1
HIF1A
FOS
MYBL2
JUN
ATF4
NFE2L2
MYC
rank 1
rank 2
rank 2
rank 4
rank 5
rank 6
rank 7
rank 8
rank 9
rank 9
rank 45
rank 92
rank 147
rank 212
0.0
0.2
0.4
0.6
0.8
1.0
De-overlapped ρ with hypoxia
b
All patients
GSE131907
GSE274934
ρ=0.87 [0.65, 0.95]
n=16; P=1.3×10−5
ρ=0.70 [0.17, 0.92]
n=11; P=0.016
ρ=0.50 [-0.68, 0.96]
n=5; P=0.391
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
scRNA patient-level ρ
c
ρ=0.307
Holm P=2.8×10−23
TCGA Hallmark hypoxia score
TCGA DoRothEA BACH1 score
LUAD
LUSC
d
Combined
LUAD
LUSC
0.314 [0.245, 0.383]
P=4.3×10−19
0.271 [0.116, 0.426]
P=5.9×10−4
0.264 [0.166, 0.363]
P=1.4×10−7
Adjusted standardized beta